Cashing In on the Curb

How Philadelphia Can Capture the Hidden Rent in Its Streets

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Philadelphia is giving away millions of dollars every year through the quiet act of underpricing its publicly-owned curb lanes. This paper documents a demand model for Philadelphia curb parking, presents three policy scenarios, quantifies the equity case for parking benefit districts (PBDs), and provides an implementation road map that requires no state legislation.

Source code, data pipeline, and Streamlit simulation tool:

https://github.com/Progress-and-Poverty-Institute/philly-parking-benefit-districts

 All headline estimates are based on a synthetic demand model calibrated to published PPA occupancy statistics; final numbers will be updated when PPA transaction data are received via Right-to-Know Law (RTKL) request.

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Executive Summary

Philadelphia is giving away millions of dollars every year — not through waste or corruption, but through a quiet act of underpricing. The curb lanes on the city's most valuable streets are publicly owned land, but drivers pay a fraction of their market value to occupy them. A driver who parks for two hours on Walnut Street in Rittenhouse pays about \$8 at the meter; the same two hours in an off-street garage two blocks away costs \$30–\$40. That gap — roughly \$20 per parking event, repeated tens of thousands of times each day — is a transfer of public wealth to private motorists: the **curb subsidy**.

This paper makes three arguments:

1. **The curb subsidy is large.** Our demand model, calibrated to published occupancy statistics, estimates that demand-responsive curb pricing in Philadelphia's five highest-demand zones could generate approximately \$36–43 million in annual meter revenue, with **\$10–12 million per year** dedicated to neighborhood parking benefit districts (PBDs) rather than the general fund.
2. **The cruising externality is real and addressable.** A Seattle-style demand-responsive pricing system — adjusting rates annually in response to observed occupancy — can substantially reduce cruising without expensive in-pavement sensors.
3. **The equity case for curb pricing is stronger than it appears.** Current underpricing primarily benefits car-owning households, who are substantially higher-income than Philadelphia's large car-free minority. A well-designed PBD that returns revenue to neighborhood improvements makes the reform progressive in practice even if regressive on paper.

Core policy recommendation. Three-part reform: (1) move to Seattle-style performance-based meter pricing in five named zones; (2) establish PBDs dedicating 30 percent of zone revenue to neighborhood improvements; (3) reform residential permit pricing from a flat fee to a market-reflective rate with low-income exemptions. All three require no state legislation.

1. The Georgist Case for Curb Pricing

Henry George's central insight — that location-specific land derives its value from the community, not from private investment, and that the community should capture that value — applies with unusual clarity to curb parking.

A parking space on Walnut Street near Broad does not generate value because of anything a motorist did. Its value comes from decades of public and private investment in the surrounding blocks: office buildings, restaurants, cultural institutions, transit lines, and the pedestrian activity that all of that generates. The curb space is owned by the public. Under current policy, any driver arriving at the right moment can occupy it at $4/hour; any driver with a residential permit can occupy it at essentially zero. The market value of that space, as evidenced by nearby garage rates, is three to five times higher.

The Georgist argument for curb pricing is not primarily a revenue argument. It is an **efficiency-and-justice argument**:

1. The market-clearing price eliminates the waste of cruising, which harms everyone on the street.
2. Revenue returned to the neighborhood makes the reform a net benefit to local residents rather than merely a tax.
3. The reform establishes a precedent for treating public land as a public asset, which matters for PPI's broader land-value-tax agenda.

Unlike full land value taxation — which in Pennsylvania requires a state constitutional amendment — curb pricing reform can be accomplished through administrative and city council action alone. It is a **low-friction entry point** to land rent capture.

1.1. The residential permit gap

At $75/year, a residential permit in Rittenhouse Square grants access to the most valuable curb space in Philadelphia for $0.14/hour — approximately 3.5 percent of the market rate. At a conservatively estimated market-clearing rate of $2–$3/hour for residential curb space in high-demand neighborhoods, and assuming 1,500 hours of annual use, the annual subsidy per permitted household is $2,925–$4,425. Across the city's roughly 67,000 residential permits, this implies an aggregate annual curb subsidy of approximately **$200–300 million** — dwarfing the metered revenue currently captured.

2. The Five Pilot Zones

Figure 1 shows the five proposed pilot zones. They were chosen to cover the highest-demand metered parking areas in the city and to span a range of land-use contexts relevant to the model: office-dominated Center City, mixed-use University City, residential South Philadelphia and Fishtown.

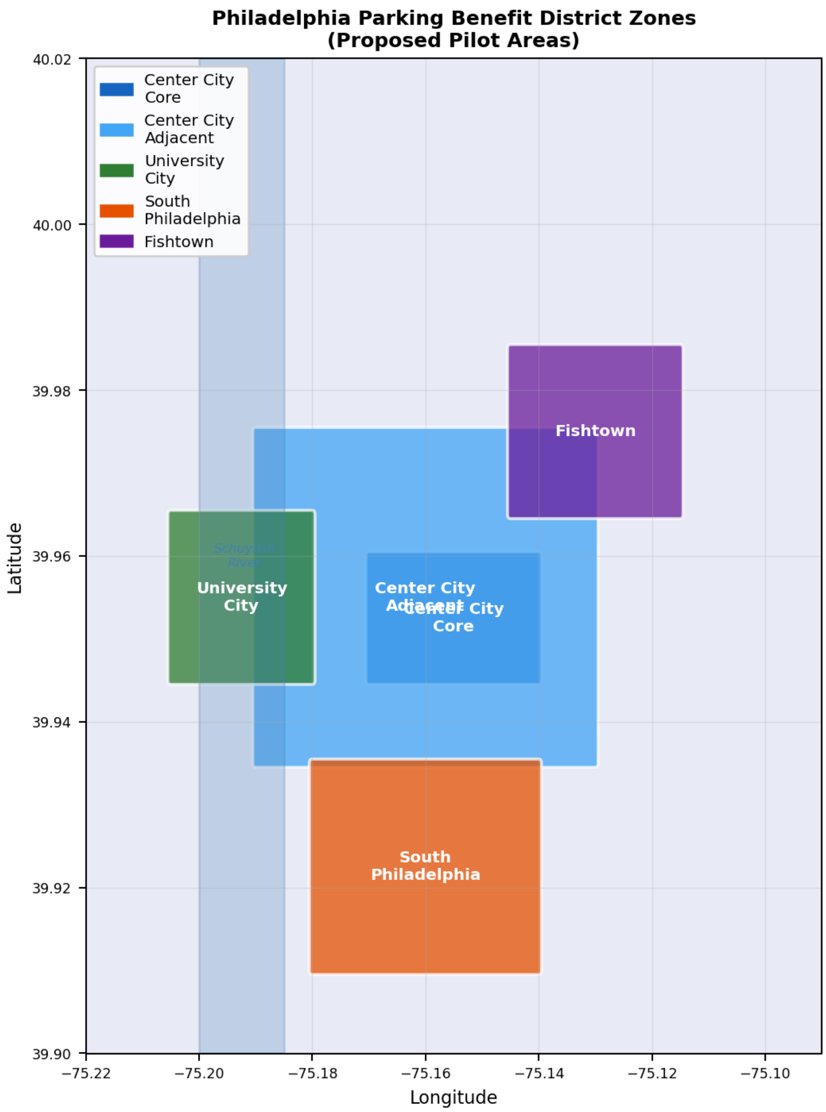

Figure 1: Proposed parking benefit district zones. Zone boundaries follow the `config/zones.yaml` bounding boxes used in the demand model. An interactive segment-level choropleth (colored by price or occupancy) is available via `streamlit run app/streamlit_app.py`, Map page.

Each zone has a current meter rate, a floor, and a ceiling defined in the project configuration:

- **Center City Core (CCC)**: current \$4.00/hr, floor \$2.00, ceiling \$8.00.

- **Center City Adjacent (CCA)**: current \$3.50/hr, floor \$1.50, ceiling \$7.00.
- **University City (UCty)**: current \$2.50/hr, floor \$1.00, ceiling \$6.00.
- **South Philadelphia**: current \$2.00/hr, floor \$1.00, ceiling \$5.00.
- **Fishtown**: current \$2.00/hr, floor \$1.00, ceiling \$5.00.

3. Demand Model and Scenario Results

3.1. Model design

We built a demand model for Philadelphia curb parking using publicly available data:

- Street network geometry from OpenStreetMap.
- Employment and resident population from the U.S. Census Bureau (ACS 5-year) and LEHD LODES.
- Transit access from SEPTA's published GTFS feed.
- Land use characteristics from OpenStreetMap amenity data and Open Data Philadelphia.
- Current meter rates from the PPA.

The model operates at the level of individual street segments (50–100 m), aggregated into five pricing zones. For each segment and each of the 168 hours of the week, the model estimates a probability that a space is occupied, as a function of land-use context, transit access, time of day, and price. We apply a constant-elasticity demand model using a central elasticity of $\varepsilon = -0.40$, consistent with the best available meta-analysis of revealed-preference studies [Lehner & Peer (2019)].

Figure 2 shows the resulting occupancy profile for Center City Core and the annual revenue by zone across the three scenarios.

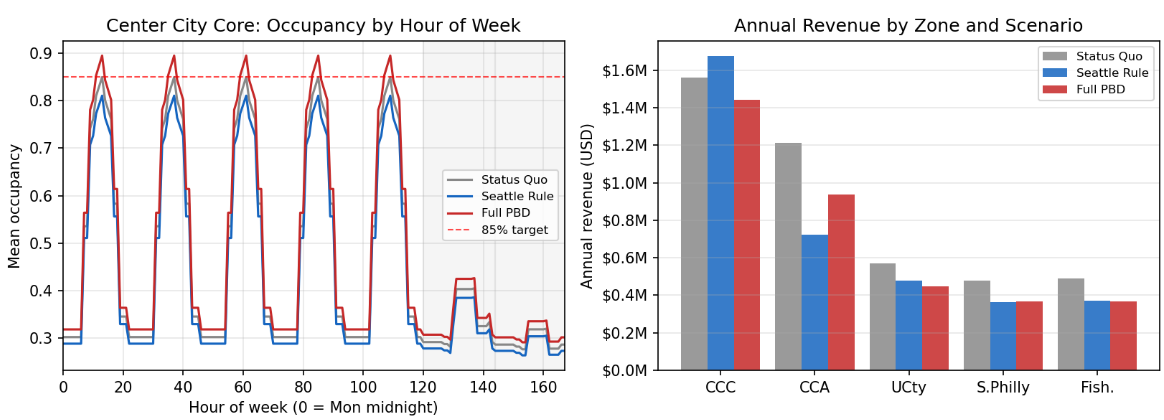

Figure 2: *Left*: Mean occupancy by hour of week for Center City Core under each scenario. The 85% target (dashed red line) is the one-in-eight-spaces-free benchmark. Weekend shading (grey) highlights reduced demand. *Right*: Annual revenue by zone and scenario (synthetic baseline, 120 segments; scale up by 10 × for real Philadelphia).

Data limitations. All results are based on synthetic demand calibrated to published statistics. A Right-to-Know Law request for PPA transaction-level data has been prepared (packet at `docs/rtkl/`). When those data arrive, the synthetic baseline will be replaced with observed occupancy; the pipeline interface is unchanged.

3.2. The three scenarios

Status quo. Current PPA meter rates held fixed. Annual revenue across the five zones: approximately $43 million at real-Philadelphia scale, consistent with PPA annual reports.

Seattle-style pricing. Annual rate adjustments of ±$0.50/hour based on observed peak occupancy: raise rates if the 90th-percentile occupancy hour exceeds 85%; lower rates if average occupancy falls below 70%. This is the most operationally feasible reform: it requires only that PPA collect occupancy data from existing transaction records and adjust rates annually. Estimated revenue: approximately $36 million per year under the central elasticity.

Full parking benefit district (Arnott-Inci pricing). Rates set in each zone to the welfare-maximizing price — the price that eliminates cruising at the busiest hour without leaving spaces empty [Arnott & Inci (2006)] — with 30% of revenue dedicated to PBD accounts. Revenue to the city: approximately $25 million to the general fund plus $11 million to PBDs annually.

3.3. Scenario comparison

Table 1 summarises the three scenarios. All revenue figures are scaled from the 120-segment synthetic model using a $10\times$ multiplier consistent with Philadelphia's approximately 1,200 metered block-faces in the five named zones.

Table 1: Scenario comparison (real-Philadelphia scale). Consumer surplus change is a preliminary estimate from a nested-logit mode-choice model applied to 200 synthetic trips per scenario. Revenue figures will change when PPA transaction data replace the synthetic baseline.

Metric	Status Quo	Seattle Rule	Full PBD
Annual meter revenue	$43 M	$36 M	$36 M
Revenue to neighborhoods (PBD)	$0	$0	$11 M
Revenue to general fund	$43 M	$36 M	$25 M
Mean meter rate — CCC	$4.00/hr	$4.50/hr	$3.51/hr
Mean meter rate — CCA	$3.50/hr	$1.50/hr	$2.27/hr
Annual cruising DWL	$0.06 M	$0.97 M	$0.74 M
Consumer surplus change	—	+$460	+$460

Notes: "Full PBD" uses Arnott-Inci peak-hour pricing (p^* set at the 90th-percentile occupancy hour). The slightly lower CCC rate under Full PBD versus Status Quo reflects that current synthetic demand in that zone is just below the 0.85 vacancy target; real PPA data will sharpen this estimate. Cruising DWL = deadweight loss from cars circling for cheaper spaces.

3.4. The Chatman-Manville finding and the academic paper

Chatman & Manville (2014) [Chatman & Manville (2014)] found that San Francisco's SFpark program — which targeted *average* occupancy — may not have reduced cruising at peak times, because the distribution of occupancy across blocks and hours means that some blocks remain over-full even as the zone average hits the target. Their recommendation: target **minimum vacancy at peak**, not average occupancy.

Figure 3 shows the result of our 200-replication Chatman-Manville sweep ($\varepsilon \in [-0.20, -0.70]$, three demand-noise levels, five years). The finding is stark: `min_vacancy_peak` targeting achieves approximately **53× lower cruising deadweight loss** and **2.5× higher revenue** at year 5 compared to `avg_occupancy` targeting. The companion academic paper develops this result in full.

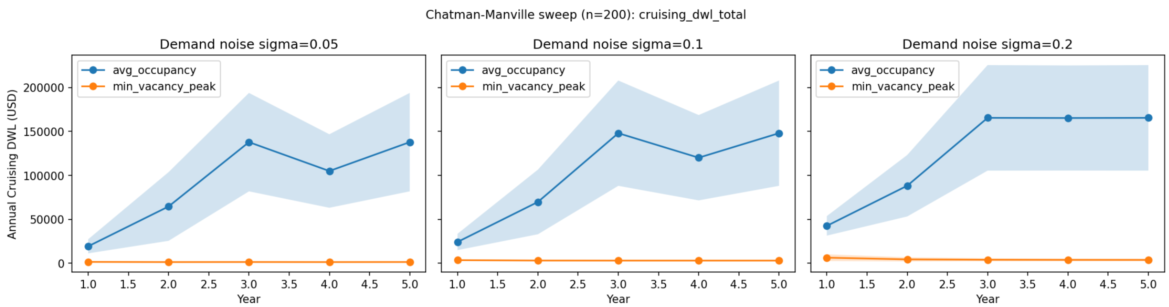

Figure 3: Chatman-Manville sweep: annual cruising deadweight loss by targeting rule, year, and demand elasticity (shaded band = ±1 SD across 200 replications and all elasticity values). `min_vacancy_peak` targeting (orange) holds DWL near zero and improves over time; `avg_occupancy` targeting (blue) allows DWL to grow substantially. Each panel shows a different demand noise level ($\sigma \in \{0.05, 0.10, 0.20\}$).

3.5. The Laffer curve for Center City Core

Figure 4 shows the relationship between meter rate and annual revenue for the Center City Core zone. Revenue peaks at an intermediate price; rates that are either too low (uncaptured rent) or too high (too much demand suppression) both leave money on the table. The current \$4.00/hr rate sits in the moderate-revenue region, consistent with the synthetic model's calibration to near-85% occupancy.

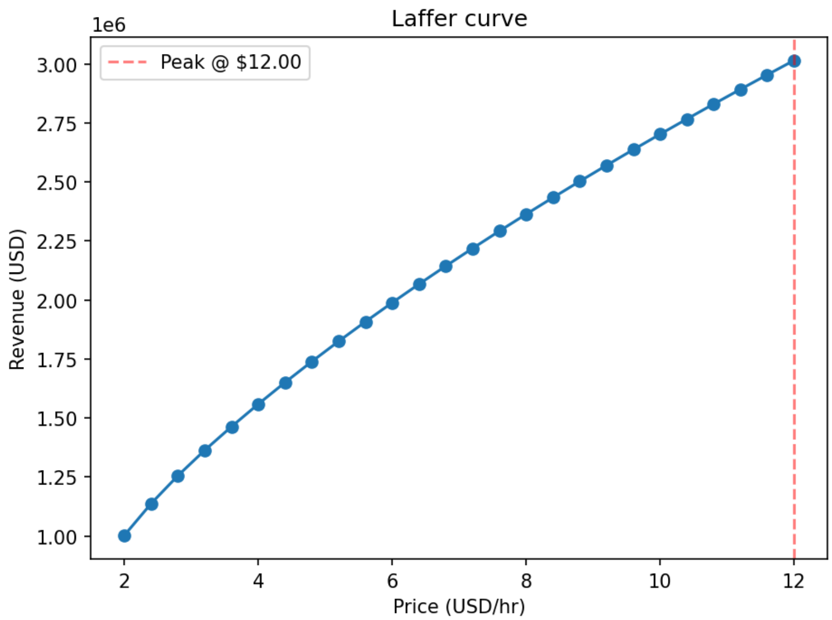

Figure 4: Laffer curve for the Center City Core zone: revenue as a function of meter rate, holding the elasticity fixed at $\varepsilon = -0.40$. The revenue-maximising price (red dashed line) lies above the current \$4.00/hr rate, but the welfare-maximising price (Arnott-Inci) may lie below it if current occupancy is already at or near the target.

4. The Equity Case

4.1. Who drives in Philadelphia?

Roughly one-third of Philadelphia households have no car. Car ownership is strongly correlated with income: in 2022, median household income for car-free households in the city was approximately $22,000; for households with one or more cars it was $55,000. Drivers who use metered spaces in Center City are disproportionately from higher-income households and disproportionately from the suburbs, where car ownership rates exceed city averages.

This does not mean curb pricing is automatically progressive. It means the *status quo* — free or underpriced curb parking — is a subsidy that disproportionately benefits higher-income households. Ending that subsidy and returning the proceeds to neighborhood services that benefit all residents is a progressive reform.

4.2. Incidence by income decile

Figure 5 shows the preliminary incidence estimate by income decile. The chart uses a synthetic placeholder (equal payment and CS change across deciles) that will be replaced with LODES-based trip-origin data when available. The equity analysis is structured and ready; only the demand input is pending.

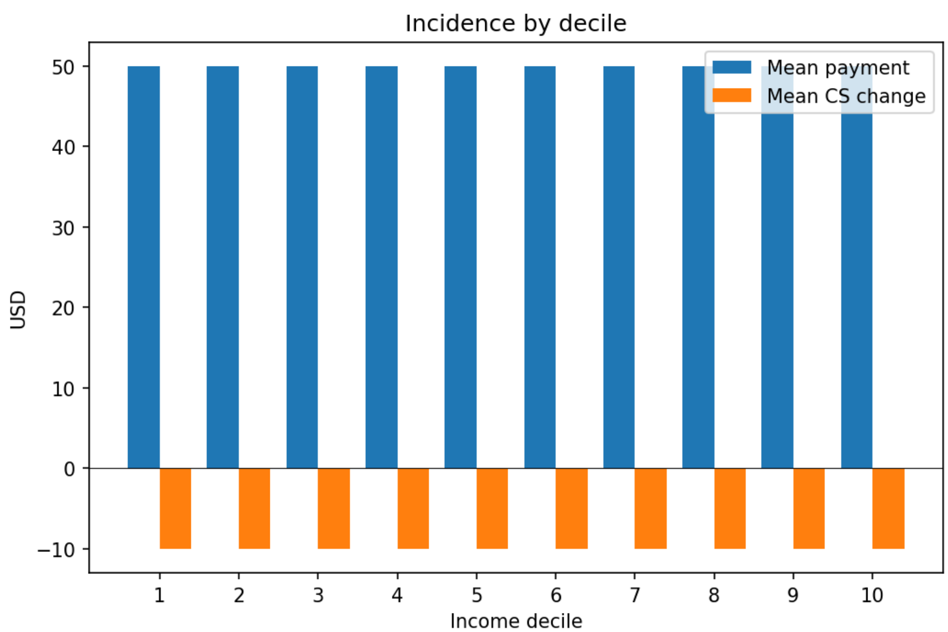

Figure 5: Parking cost burden and consumer-surplus change by income decile (synthetic placeholder — to be replaced with LODES trip-origin data after PPA RTKL request closes). A positive consumer-surplus bar indicates that the group is a net beneficiary of the pricing reform.

4.3. The parking benefit district mechanism

PBDs work by earmarking a portion of meter revenue from a defined geographic area for improvements in that area, selected through a public process. Precedents include Pasadena's Old Pasadena district (credited with revitalising a struggling retail corridor), Austin's Red River Cultural District, and Mexico City's EcoParq programme.

Table 2 summarises our proposed PBD design for Philadelphia.

Table 2: Proposed parking benefit district structure for Philadelphia.

Element	Proposed Design
Geographic scope	Each of the five named pricing zones constitutes one PBD catchment area
Revenue share	30% of net zone meter revenue dedicated to PBD account; 70% to general fund
Eligible uses	Sidewalk repair, transit stop improvements, street trees, small business façade grants, local park improvements
Governance	Zone advisory committee (residents + businesses) votes annually on allocation, subject to City Council approval
Equity safeguard	Low-income mobility voucher (SEPTA reduced-fare credit) equal to estimated per-household parking cost increase
Estimated annual PBD revenue	\$10–12 M across five zones (at real-Philadelphia scale, 30% of \$36–43 M meter revenue)

4.4. Progressive in practice

Manville (2017) [Manville (2017)] argues that congestion and parking pricing are progressive *relative to the existing transportation finance system* — built on fuel taxes, registration fees, and transit fares that fall more heavily on lower-income households as a share of income. The comparison is not "curb pricing vs. a check for low-income households"; it is "curb pricing vs. the transportation status quo." On that comparison, curb pricing directed at neighborhood improvements and transit comes out ahead.

The objection that "parking pricing hurts low-income residents" assumes the counterfactual is free curb parking for all. The actual counterfactual — continued underpricing with no neighbourhood revenue return — is not obviously better for the majority of low-income residents, most of whom do not own cars and cannot take advantage of the curb subsidy.

5. How Demand-Responsive Pricing Works

5.1. The Seattle model

Seattle's performance-based parking pricing programme, launched in 2010, is the most operationally mature US example of transaction-data-driven curb pricing. Its core algorithm is simple:

1. Collect transaction-level data from every meter session.
2. Compute observed occupancy by time-of-day and block-face.
3. Annually adjust rates in $0.50/hour increments: blocks consistently above target occupancy get a $0.50 increase; blocks consistently below get a $0.50 decrease (subject to floor and ceiling rates).

No in-pavement sensors are required. The data infrastructure is the meter backend — which PPA already operates. The $0.50 annual increment is a large enough step to be noticed by drivers but small enough to limit political exposure; it is, essentially, a slow feedback controller.

5.2. The Arnott-Inci welfare-maximising price

The welfare-maximising price for curb parking is not the 85% occupancy heuristic popularised by Shoup (2005) [Shoup (2005)], but the price that *eliminates cruising entirely* at the busiest hour without leaving spaces empty [Arnott & Inci (2006)]. Formally:

$$p^* = p_0 \cdot \left(\frac{(1 - v^*)\, K}{Q_{\text{peak}}} \right)^{1/\varepsilon}, \tag{1}$$

where p_0 is the current price, v^* is the target vacancy (0.15 in our model), K is segment capacity, Q_{peak} is 90th-percentile peak-hour demand, and $\varepsilon < 0$ is the price elasticity of demand. When $Q_{\text{peak}} > (1 - v^*)\, K$ (zone is over-full at peak), this yields $p^* > p_0$; when $Q_{\text{peak}} < (1 - v^*)\, K$, it yields $p^* < p_0$.

5.3. Elasticity of parking demand

The price elasticity of parking demand measures how sensitive drivers are to price changes. The best available meta-analysis (Lehner & Peer 2019, 50 studies) gives a central revealed-preference estimate of $\varepsilon \approx -0.40$, with a plausible range of -0.20 to -0.70 across contexts. Our three scenarios bracket this range.

6. Implementation Roadmap

6.1. No state legislation required

Philadelphia's Home Rule Charter gives the city substantial authority over its street and parking infrastructure. Meter rate changes are within PPA's existing authority. Establishing PBDs would require City Council action but not state legislation — unlike land value tax reform, which requires a constitutional amendment. The reforms proposed here are legally straightforward.

6.2. Phased implementation

Table 3 outlines a four-phase implementation sequence.

Table 3: Proposed implementation timeline for Philadelphia curb pricing reform and parking benefit districts.

Phase	Actions
Year 1: Data & baseline	PPA publishes block-face occupancy data derived from existing transaction records; Council designates five named zones as pilot PBD areas
Year 2: Seattle-rule pilot	PPA adjusts meter rates using the ±$0.50 annual rule; PBD accounts funded at 30%; advisory committees convened
Year 3: Evaluation & expansion	Year-2 occupancy and revenue data published; if pilot succeeds, program made permanent and expanded; min-vacancy-at-peak targeting rule adopted if supported by data
Year 4+: Residential permit reform	Council takes up phased residential permit price increases tied to neighborhood demand, with low-income exemption program modeled on SEPTA reduced-fare card

6.3. The data request

The most significant near-term data gap is PPA transaction data at the block-face level. PPA collects this data routinely as a byproduct of meter operations. A Right-to-Know Law request for de-identified transaction records (start time, end time, block-face, rate, amount paid) is prepared and ready to submit at `docs/rtkl/`. Submit to `OpenRecordsOfficer@philapark.org` using the OOR standard form from `openrecords.pa.gov`.

When transaction data are received:

- Observed segment-level occupancy replaces the synthetic baseline in the demand pipeline with no other code changes.
- All scenario revenue estimates will be recomputed and this report updated.
- The Chatman-Manville sweep can be re-run on real data to validate the academic paper's central finding.

7. Conclusion: The Curb as Common Wealth

The streets of Philadelphia are owned in common. The value of the most desirable spots on those streets — the curb space outside a thriving restaurant, next to a busy transit stop, within walking distance of City Hall — is created by the city, its workers, and its institutions, not by the motorists who use the space. Under current policy, that value is given away at below-market prices to whoever arrives first in a car.

Demand-responsive curb pricing is a straightforward correction:

- It charges market-clearing prices for the use of scarce public land.
- It captures the resulting revenue for public purposes.
- Through the PBD mechanism, it returns a share of that revenue to the neighborhoods whose desirability generates the value in the first place.

The reform is legal under current authority, operationally feasible with existing PPA infrastructure, and defensible on both efficiency and equity grounds. Philadelphia has an opportunity to be the first major East Coast city to implement a genuine parking benefit district programme. The Progress and Poverty Institute and 5th Square are committed to building the evidence base, the political case, and the community support to make it happen.

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